

HOPEDALE, NL, Canada

WMO: 719000

Lat: 55.449N

Lon: 60.228W

Elev: 39

StdP: 14.68

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.1	-14.8	-28.1	1.1	-16.4	-24.4	1.4	-13.3	42.5	7.8	38.9	5.8	14.3	270	0.900

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	71.9	60.1	67.8	58.3	64.2	56.5	61.3	68.8	59.3	66.2	57.4	63.4	13.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.9	71.7	64.7	55.8	66.5	63.4	53.9	61.8	61.3	27.4	69.3	26.0	66.3	24.8	63.4	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.8	30.7	27.0	DB	-22.2	81.8	3.7	3.0	-24.8	83.9	-27.0	85.7	-29.0	87.4	-31.7	89.5	
			WB	-21.7	66.2	3.7	2.1	-24.4	67.7	-26.5	68.9	-28.5	70.1	-31.2	71.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.9	3.8	2.7	12.5	24.8	35.1	44.0	52.5	54.1	47.5	38.3	27.1	14.5
	DBStd	19.51	12.37	10.59	11.50	8.76	5.72	7.92	7.43	6.51	6.08	5.40	6.84	10.75
	HDD50	7721	1433	1324	1163	755	464	220	61	30	121	365	686	1099
	HDD65	12822	1898	1744	1628	1205	927	632	392	342	526	828	1136	1564
	CDD50	385	0	0	0	0	2	40	138	158	45	2	0	0
	CDD65	10	0	0	0	0	0	1	4	5	0	0	0	0
	CDH74	75	0	0	0	0	0	14	21	37	3	0	0	0
	CDH80	10	0	0	0	0	0	2	1	7	0	0	0	0
	WSAvg	12.4	14.7	13.8	13.4	12.1	10.3	10.3	9.5	9.8	12.0	12.9	14.7	15.2
	PrecAvg	34.8	3.3	3.3	3.3	2.4	2.2	2.6	3.5	2.8	2.4	2.8	2.8	3.0
(15) Precipitation	PrecMax	45.6	6.3	7.7	8.5	7.3	5.0	6.7	7.8	7.6	5.2	5.4	4.9	6.5
	PrecMin	20.4	0.8	0.2	0.5	0.4	0.7	0.6	1.1	0.7	0.6	1.1	0.6	1.0
	PrecStd	6.3	1.6	2.1	2.0	1.5	1.0	1.7	1.9	1.7	1.1	1.2	1.2	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	33.7	40.3	46.7	60.4	75.8	76.7	79.0	70.5	57.6	46.0	37.5
		MCWB	34.2	32.6	34.1	39.7	49.6	61.1	61.1	64.2	57.3	50.9	42.4	35.8
	2%	DB	32.2	29.3	35.2	42.4	51.3	67.4	71.3	71.8	65.2	52.1	40.7	34.7
		MCWB	31.4	28.8	32.6	37.3	43.3	56.6	60.0	60.8	55.8	46.5	37.5	33.8
	5%	DB	29.2	24.8	33.0	39.4	46.6	61.4	67.1	67.4	60.9	48.7	37.8	32.5
		MCWB	28.5	23.9	31.5	35.7	40.9	53.1	58.4	58.9	53.8	44.6	35.4	31.9
	10%	DB	24.7	19.3	30.5	36.2	42.9	55.7	63.4	63.7	56.9	45.5	35.7	30.0
		MCWB	24.0	18.2	29.0	33.4	38.5	49.1	56.5	56.8	50.5	41.6	34.0	29.2
	0.4%	WB	34.4	32.8	34.6	40.5	49.6	62.6	63.8	64.6	60.3	52.1	41.3	36.2
		MCDB	34.7	33.5	36.1	45.9	58.3	73.9	71.4	72.8	65.6	56.2	43.1	37.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.5	28.3	32.8	38.2	44.0	57.7	61.1	61.5	57.0	48.1	37.1	34.0
		MCDB	32.0	28.7	33.9	42.4	49.1	66.3	68.8	68.4	63.3	52.3	38.6	34.7
	5%	WB	28.7	22.8	31.2	36.3	41.0	53.4	59.4	59.3	54.4	45.1	35.5	32.1
		MCDB	29.3	23.5	32.4	39.6	46.2	60.7	66.6	65.4	61.0	48.7	36.8	32.8
	10%	WB	24.4	18.3	28.4	34.1	38.7	49.5	57.4	57.3	51.4	42.3	34.0	29.3
		MCDB	25.2	19.4	29.8	36.2	42.7	55.3	63.4	62.7	57.1	45.4	35.1	30.0
	5% DB	MDBR	12.2	13.6	14.0	12.3	10.6	11.8	12.5	11.9	10.6	7.4	7.6	9.8
		MCDBR	14.0	18.8	14.5	16.1	18.7	22.9	19.7	19.6	18.0	13.1	9.8	9.4
		MCWBR	11.8	19.3	11.4	13.2	12.4	13.7	11.1	10.3	11.1	10.2	9.3	9.1
		MCWBR	12.1	18.8	11.9	15.4	17.6	22.1	18.6	17.1	17.4	13.2	8.6	9.1
(38) Clear-Sky Solar Irradiance	taub		0.239	0.265	0.277	0.338	0.375	0.377	0.400	0.370	0.344	0.306	0.259	0.243
		taud	2.228	2.213	2.204	2.071	2.156	2.338	2.358	2.443	2.471	2.525	2.357	2.162
	Ebn at Noon		224	259	285	278	274	275	266	268	260	246	217	187
		Edh at Noon	17	26	33	44	44	37	36	31	26	20	15	15
(44) All-Sky Solar Radiation	RadAvg		236	531	990	1473	1717	1610	1450	1222	912	475	239	158
	RadStd		26	67	122	107	177	157	123	101	73	30	23	16

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page